

Financial Customer Intelligence & Risk Analytics

Customer Behavior, Transaction Patterns & Payment Risk Analysis



Prepared by:

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Data Analyst

Tools Used:

Excel | Power Query | SQL | Power BI

Dataset:

Financial Customer Intelligence & Risk Analytics

85,000 Records | 4 Related Tables

Project Overview

This project analyzes **customer profiles, card portfolios, transaction activity, and payment behavior** to generate actionable insights into customer engagement, spending patterns, and potential financial risk.

The analysis uses four related datasets — **Customers, Cards, Transactions, and Payments** — containing **85,000 records**. The project follows an end-to-end analytics workflow, beginning with data profiling and validation in **Power Query**, followed by business-focused analysis using **MySQL and SQL**. **Excel** is used to create supporting analysis and visualizations, while **Power BI** is used to develop an interactive dashboard for monitoring key customer, transaction, and payment metrics.

The primary objective is to understand **who the customers are, how they use their cards, where transaction activity is concentrated, and how payment behavior varies across customer segments**. The findings are then translated into business insights and recommendations that can support **customer engagement, portfolio management, payment-risk monitoring, and data-driven decision-making**.

Project Scope

- **Customer Intelligence** — customer demographics, income, credit profile, and segmentation
- **Card Analytics** — card types, credit limits, card activity, and portfolio behavior
- **Transaction Analytics** — transaction volume, value, categories, trends, and spending patterns
- **Payment & Risk Analytics** — payment status, payment behavior, days past due, and risk indicators
- **Business Insights** — identifying patterns and converting analytical findings into actionable recommendations

Analytical Workflow

Power Query → MySQL/SQL → Excel → Power BI → Business Insights & Recommendations

Business Problem

Financial institutions manage large volumes of customer, card, transaction, and payment data. However, analyzing these datasets together is essential to understand **customer value, spending behavior, card usage, and payment patterns**

The objective of this project is to transform raw financial data into **actionable business intelligence** by answering key questions across customer engagement and payment-risk areas.

Key Business Questions

Customer Intelligence

- Which customer segments generate the highest transaction value?
- How does income and credit profile relate to customer spending behavior?
- Which customers demonstrate high-value or highly active behavior?

Card Portfolio

- Which card types have the highest customer adoption and transaction activity?
- How does credit-limit distribution vary across card types?
- Which card categories contribute most to overall transaction value?

Transaction Analytics

- What are the major transaction categories and spending patterns?
- How does transaction activity change over time?
- Which customer, card, or merchant segments contribute the most transaction value?
- What proportion of transactions are approved, declined, or reversed?

Payment & Risk

- What proportion of payments are made on time, late, partially, or missed?
- Which customer segments show higher payment delays?
- Which customers demonstrate repeated late or missed payment behavior?
- How does payment behavior vary by card type or customer profile?

Business Objective

Identify meaningful patterns in customer engagement, card usage, transaction activity, and payment behavior to support data-driven customer strategies, portfolio management, and risk monitoring.

Dataset & Data Model

Dataset Overview

The project uses a **synthetic financial customer dataset** containing **85,000 records across four related tables**. The dataset is designed to simulate customer, card, transaction, and payment activity for analytical and risk-monitoring purposes.

Table	Records	Key Identifier	Description
Customers	10,000	Customer_ID	Customer demographics, income, employment, and credit profile
Cards	12,000	Card_ID	Card type, credit limit, annual fee, status, and account information
Transactions	50,000	Transaction_ID	Customer spending, transaction category, amount, status, and location
Payments	13,000	Payment_ID	Payment amount, due amount, payment status, and days past due
Total	85,000	—	—

Data Model

The dataset follows a **relational model** where customers, cards, transactions, and payments are connected through common identifiers

Customers

|

| Customer_ID



Cards

|

| Card_ID



Transactions

Customers

|
| Customer_ID



Payments

Relationships

- **Customers → Cards:** One-to-Many — one customer can have multiple cards.
- **Cards → Transactions:** One-to-Many — one card can have multiple transactions.
- **Customers → Transactions:** One-to-Many — one customer can make multiple transactions.
- **Customers → Payments:** One-to-Many — one customer can have multiple payment records.
- **Cards → Payments:** One-to-Many — one card can have multiple payment records.

Primary Keys: Customer_ID, Card_ID, Transaction_ID, Payment_ID

Foreign Keys: Customer_ID, Card_ID

Data Source: Synthetic dataset created for analytical and portfolio purposes. It does not represent proprietary customer data from any financial institution.

Data Profiling, Cleaning & Validation

Data Profiling

The four datasets were first profiled in **Power Query** to assess their structure, completeness, uniqueness, data types, and consistency before beginning the analysis.

The following checks were performed:

- Row and column counts
- Missing and blank values
- Duplicate records and primary-key uniqueness
- Data types
- Numeric value ranges
- Categorical consistency
- Customer and card ID validity
- Relationships between the four tables

Table	Records	Missing Values	Duplicate Primary IDs	Validation Result
Customers	10,000	0	0	✓ Passed
Cards	12,000	0	0	✓ Passed
Transactions	50,000	0	0	✓ Passed
Payments	13,000	0	0	✓ Passed
Total	85,000	0	0	Validated

Data Cleaning & Preparation

The profiling process found **no material data-quality issues** requiring record-level correction or deletion. The datasets were therefore preserved without unnecessary modifications.

Power Query was used primarily to:

- Validate data quality
- Confirm identifier uniqueness
- Verify relationships between tables
- Standardize data types for analysis
- Confirm numerical and date fields were suitable for downstream analysis

01 Executive Portfolio & Financial Performance KPIs

KPI	Result
Total Customers	10,000
Total Cards	12,000
Total Transactions	50,000
Approved Transactions	45,493
Declined Transactions	3,485
Approval Rate	90.99%
Approved Transaction Value	₹100,754,644.06
Average Approved Transaction Value	₹2,214.73
Total Payment Amount	₹91,703,109.30
Total Credit Limit	₹1,864,300,000
Active Card Percentage	82.10%

Executive Insights

Insight 1 — Strong Customer and Card Portfolio

The portfolio consists of **10,000 customers and 12,000 cards**, representing approximately **1.2 cards per customer**. The portfolio provides a strong base for analyzing customer-level card ownership and engagement.

Insight 2 — High Transaction Approval Performance

The dataset records a 90.99% transaction approval rate, with 45,493 approved transactions out of 50,000. This indicates that the majority of transactions were successfully approved within the analyzed portfolio.

Insight 3 — Declined Transactions as an Optimization Area

3,485 transactions were declined, representing approximately **7.0% of total transactions**. Analyzing decline patterns across card types, merchant categories, customer segments, and time periods can help identify areas of transaction friction.

Insight 4 — Significant Credit Exposure

The card portfolio represents approximately ₹186.43 crore in aggregate credit limits. This provides a substantial base for evaluating transaction activity and customer behavior relative to available credit exposure.

Insight 5 — Meaningful Approved Transaction Activity

Approved transactions generated approximately ₹10.08 crore in transaction value, with an average approved transaction value of ₹2,214.73. This provides a foundation for identifying high-value customers, card segments, and transaction categories.

Insight 6 — Strong Active-Card Base

82.10% of cards are active, indicating that the majority of the card portfolio is currently active. The remaining cards can be further analyzed to understand inactive, blocked, or closed status patterns.

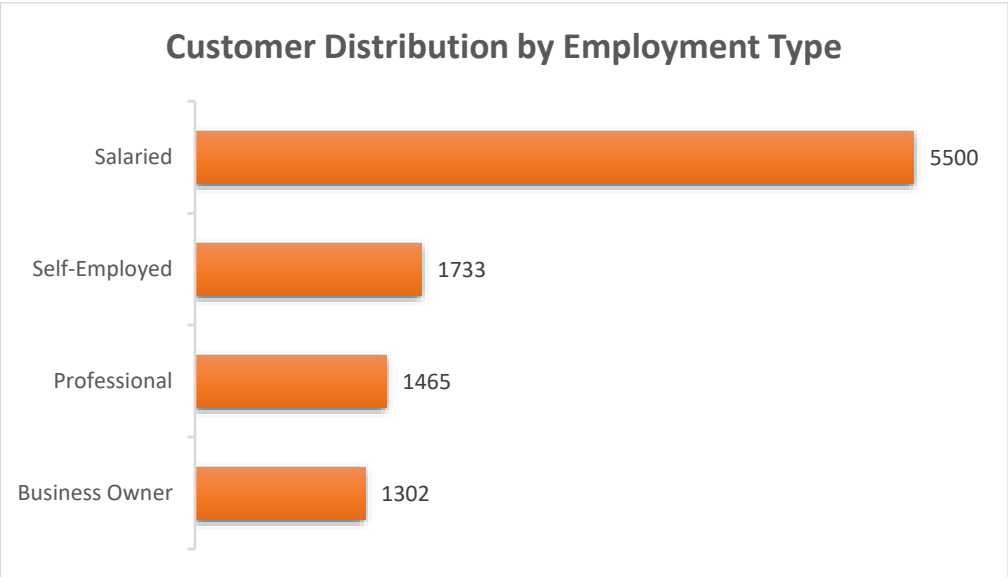
Business Recommendations

Priority Area	Recommendation
Transaction Experience	Analyze declined transactions by card type, merchant category, geography, customer segment, and time period to identify potential drivers of transaction friction.
Customer Engagement	Identify high-value and highly active customers and develop targeted engagement strategies based on transaction activity, card ownership, and customer characteristics.
Card Activation	Analyze inactive cards by customer and card characteristics to identify suitable opportunities for reactivation and improved card engagement.
Portfolio Management	Segment customers based on card ownership, transaction activity, and credit exposure to understand differences in customer behavior.
Credit Exposure	Compare transaction activity with available credit limits to understand different levels of customer credit utilization behavior.
Payment Risk Monitoring	Analyze payment status, Days Past Due, and repeated late-payment behavior to identify customer segments requiring closer monitoring.

Customer Intelligence & Segmentation

Customer Workforce Profile

Employment Type	Total Customers	Customer Distribution
Salaried	5,500	55.00%
Self-Employed	1,733	17.33%
Professional	1,465	14.65%
Business Owner	1,302	13.02%
Total	10,000	100.00%



Key Business Insight

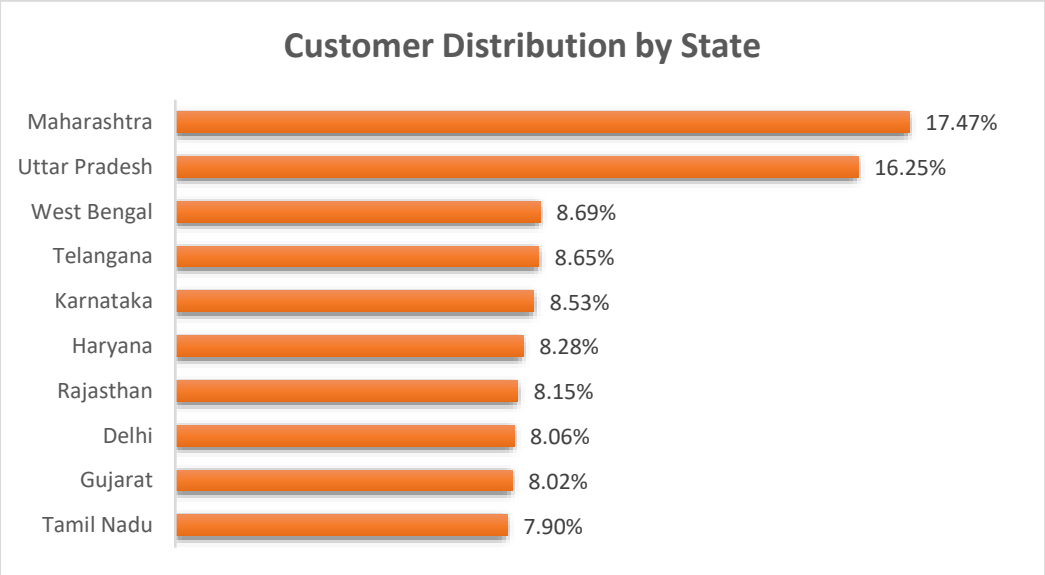
Salaried customers form the largest customer segment, accounting for 55% of the total customer base. Self-Employed, Professional, and Business Owner customers collectively represent the remaining 45%, indicating that the portfolio serves multiple employment segments but has a strong concentration among salaried customers.

Business Recommendation

Customer engagement and product strategies should prioritize the salaried segment due to its significant representation in the customer base, while tailored offerings can be developed for self-employed, professional, and business-owner segments to strengthen engagement across the broader customer portfolio.

Geographic Customer Distribution

State	Total Customers	Customer Distribution
Maharashtra	1,747	17.47%
Uttar Pradesh	1,625	16.25%
West Bengal	869	8.69%
Telangana	865	8.65%
Karnataka	853	8.53%
Haryana	828	8.28%
Rajasthan	815	8.15%
Delhi	806	8.06%
Gujarat	802	8.02%
Tamil Nadu	790	7.90%
Total	10,000	100.00%



Key Business Insight

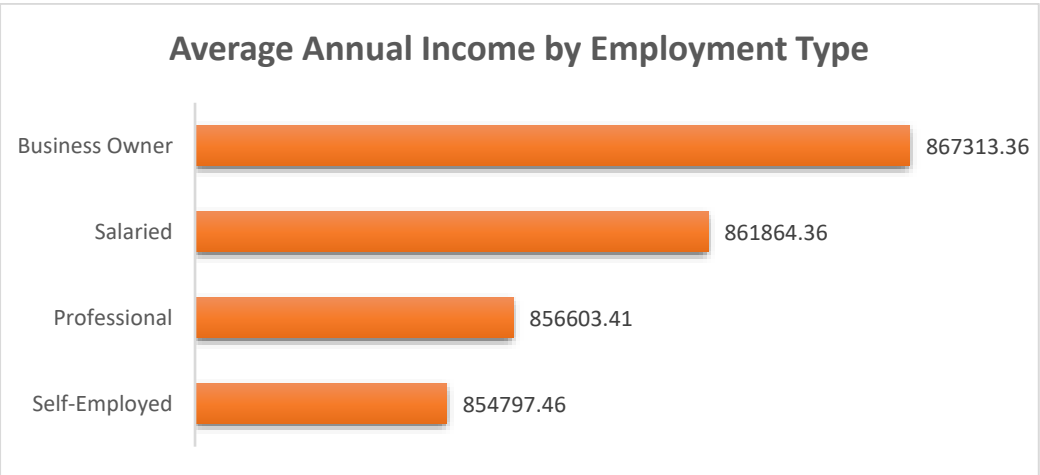
Customer distribution is concentrated in Maharashtra and Uttar Pradesh, which account for 17.47% and 16.25% of the total customer base respectively. Together, these two states represent 33.72% of customers, while the remaining customers are distributed relatively broadly across the other states.

Business Recommendation

The business can prioritize customer engagement and market-level strategies in Maharashtra and Uttar Pradesh due to their high customer concentration, while maintaining broader regional initiatives across other states to support balanced customer growth and engagement.

Customer Income Profile

Employment Type	Average Annual Income
Business Owner	₹867,313.36
Salaried	₹861,864.36
Professional	₹856,603.41
Self-Employed	₹854,797.46



Business Insight

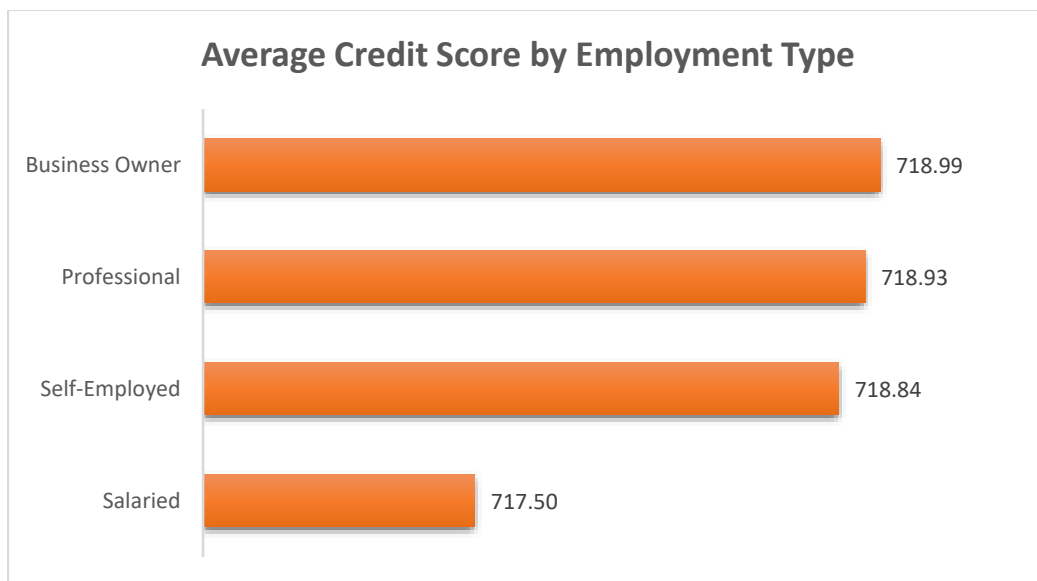
Average annual income is relatively consistent across employment segments, ranging from ₹854,797 for Self-Employed customers to ₹867,313 for Business Owners. Business Owners have the highest average income, but the relatively narrow range indicates that income levels are broadly similar across the customer segments.

Business Recommendation

The business should avoid relying on employment type alone when differentiating customers by income. Instead, employment type can be combined with credit score, card ownership, transaction activity, and payment behavior to develop more meaningful customer segments and targeted strategies.

Customer Credit Profile by Employment Type

Employment Type	Average Credit Score
Business Owner	718.99
Professional	718.93
Self-Employed	718.84
Salaried	717.50



Business Insight

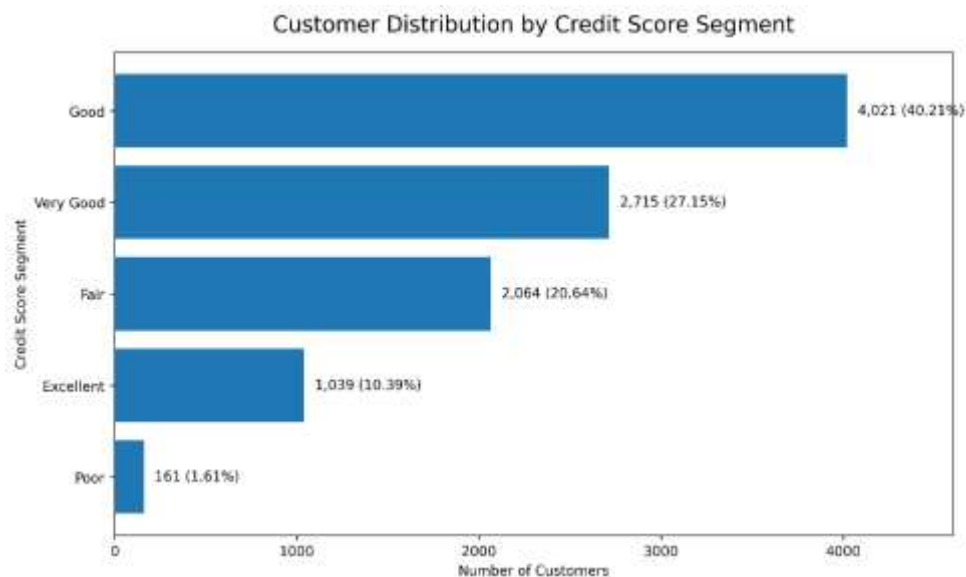
Average credit scores are highly consistent across employment types, ranging from 717.50 among Salaried customers to 718.99 among Business Owners. The narrow difference of approximately 1.5 points indicates that employment type alone does not materially differentiate the average credit profile of customers in this dataset.

Business Recommendation

Employment type should not be used as a standalone indicator for customer credit segmentation. The business should combine employment type with credit score bands, income, transaction behavior, card usage, and payment patterns to develop more meaningful customer profiles and identify potential opportunities or risk indicators.

Customer Credit Score Segmentation

Credit Segment	Total Customers	Customer Distribution
Good	4,021	40.21%
Very Good	2,715	27.15%
Fair	2,064	20.64%
Excellent	1,039	10.39%
Poor	161	1.61%
Total	10,000	100.00%



Key Business Insight

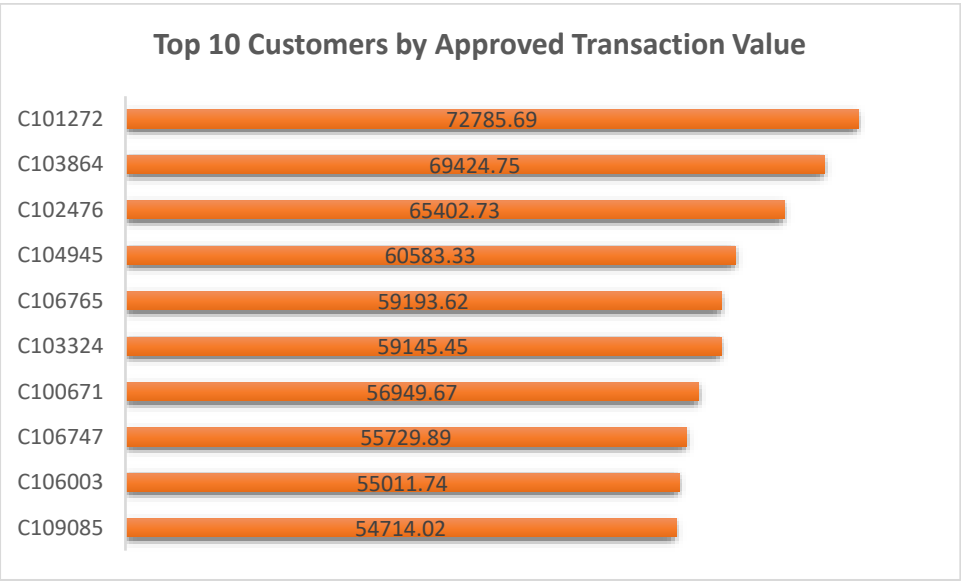
The customer portfolio is primarily concentrated within the Good and Very Good credit segments, representing 40.21% and 27.15% of customers respectively. Together, these segments account for 67.36% of the customer base, while only 1.61% of customers fall into the Poor credit segment.

Business Recommendation

The business can use credit-score segmentation as one component of customer profiling, with differentiated engagement and product strategies across credit segments. Customers in the Good and Very Good segments may represent an important target group for broader product engagement, while the smaller Poor segment can be monitored alongside payment behavior and other risk indicators before making customer-level decisions.

High-Value Customer Transaction Analysis

Rank	Customer ID	Approved Transaction Value
1	C101272	₹72,785.69
2	C103864	₹69,424.75
3	C102476	₹65,402.73
4	C104945	₹60,583.33
5	C106765	₹59,193.62
6	C103324	₹59,145.45
7	C100671	₹56,949.67
8	C106747	₹55,729.89
9	C106003	₹55,011.74
10	C109085	₹54,714.02



Key Business Insight

The top 10 customers show relatively high approved transaction spending, with **C101272** recording the highest approved transaction value of **₹72,785.69**, followed by **C103864** at **₹69,424.75**. The remaining customers in the top 10 recorded approved transaction values ranging from approximately **₹54,714 to ₹65,403**

Business Recommendation

Identify these customers as a high-value transaction segment for further analysis and targeted engagement. Evaluate their card ownership, transaction frequency, customer profile, and payment behavior to understand the characteristics associated with higher transaction activity and identify suitable opportunities for deeper customer engagement

Above-Average Customer Spending Behavior

Rank	Customer ID	Average Approved Transaction
1	C108556	₹19,424.06
2	C106372	₹16,555.01
3	C104158	₹16,295.73
4	C106831	₹15,942.44
5	C103927	₹15,829.52
6	C102476	₹15,769.09
7	C109664	₹14,234.22
8	C104692	₹14,053.04
9	C105197	₹13,667.40
10	C109761	₹12,870.41

Key Business Insight

The identified customers demonstrate substantially higher average approved transaction values than the overall approved transaction benchmark. Customer C108556 records the highest average approved transaction value at ₹19,424.06, followed by C106372 at ₹16,555.01.

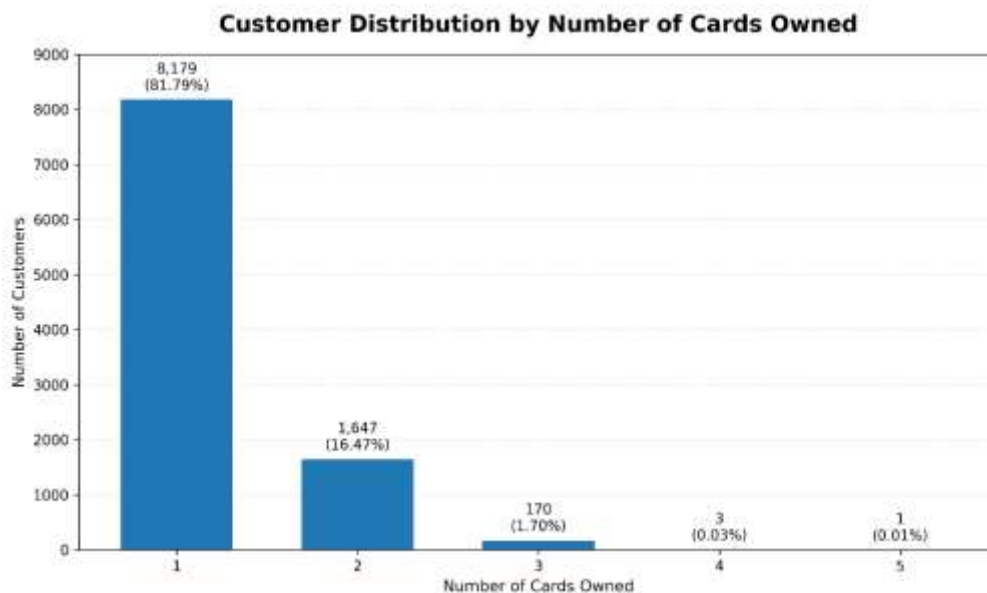
The results indicate a distinct group of customers with significantly higher average transaction values, suggesting an opportunity to further investigate their transaction frequency, card usage, income profile, and payment behavior.

Business Recommendation

The business can analyze this above-average spending group as a potential high-engagement customer segment and evaluate their broader financial behavior, including card usage, income, credit profile, and payment patterns. Insights from this analysis can support targeted customer engagement and personalized product strategies.

Card Ownership & Customer Engagement

Number of Cards	Total Customers	Customer Distribution
1	8,179	81.79%
2	1,647	16.47%
3	170	1.70%
4	3	0.03%
5	1	0.01%
Total	10,000	100.00%



Key Business Insight

The customer portfolio is predominantly composed of single-card customers, with 81.79% of customers holding one card. Customers holding two cards account for 16.47%, while only 1.74% hold three or more cards. This indicates that multi-card ownership represents a relatively small portion of the customer base

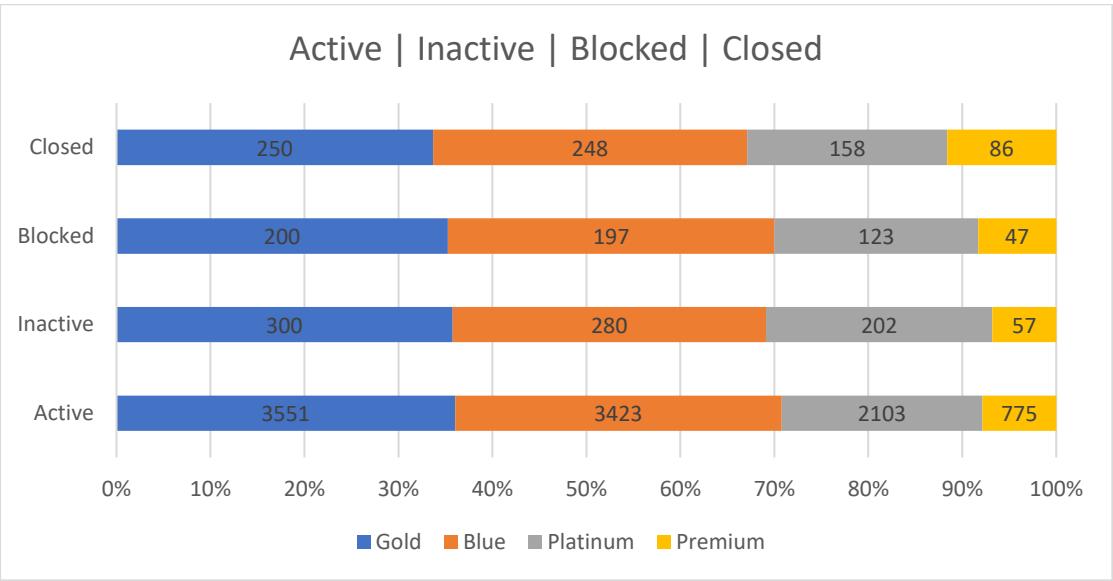
Business Recommendation

The business can investigate opportunities to increase multi-card engagement among suitable existing customers by understanding product preferences, transaction behavior, credit profiles, and payment patterns. Targeted cross-sell strategies can then be considered for customers who demonstrate a suitable profile for additional card products.

CARD PORTFOLIO & STATUS ANALYTICS

Card Portfolio Composition & Status Distribution

Card Type	Total Cards	Active	Inactive	Blocked	Closed	Active Rate
Gold	4,301	3,551	300	200	250	82.56%
Blue	4,148	3,423	280	197	248	82.52%
Platinum	2,586	2,103	202	123	158	81.32%
Premium	965	775	57	47	86	80.31%



Key Business Insight

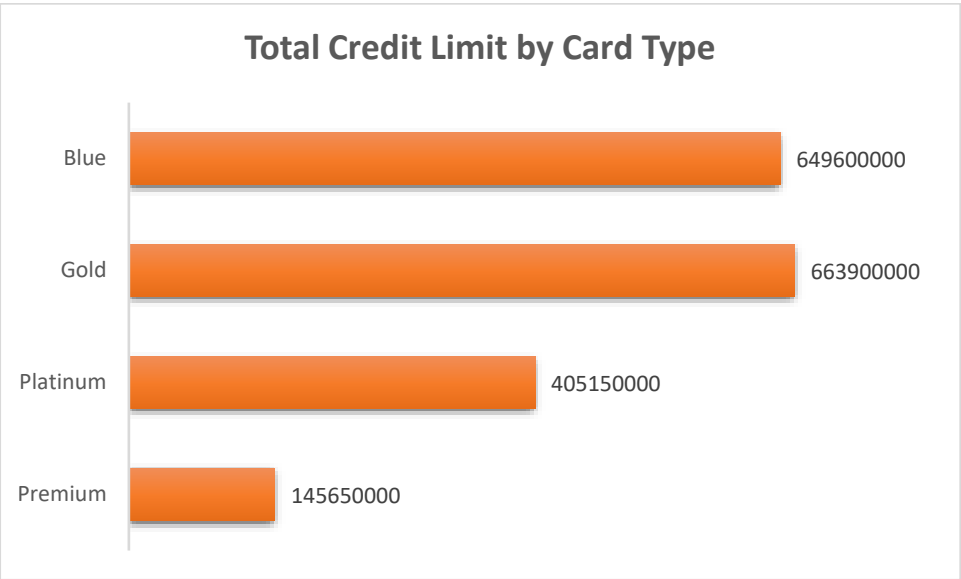
Gold and Blue cards represent the largest portions of the card portfolio, with 4,301 and 4,148 cards respectively. Their active rates are also the highest at 82.56% and 82.52%. Premium cards have the lowest active rate at 80.31%, while Platinum stands at 81.32%. Overall, 82.10% of the 12,000-card portfolio is active.

Business Recommendation

Monitor activation and status patterns across card types, particularly Premium and Platinum cards where active rates are comparatively lower. Further analysis of customer characteristics, transaction activity, and payment behavior can help identify opportunities to improve card engagement and understand the drivers of inactive, blocked, and closed statuses.

Credit Limit & Portfolio Exposure

Card Type	Average Credit Limit	Total Credit Limit
Platinum	₹156,670.53	₹405,150,000
Blue	₹156,605.59	₹649,600,000
Gold	₹154,359.45	₹663,900,000
Premium	₹150,932.64	₹145,650,000



Key Business Insight

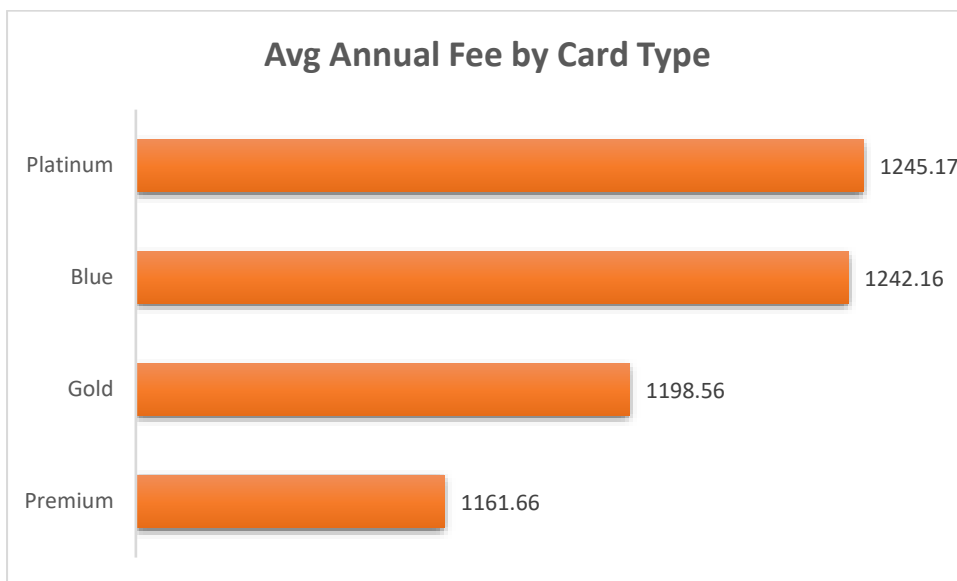
Platinum cards have the highest average credit limit at ₹156,670.53, indicating the highest average credit allocation per card among the four card types. However, Gold has the highest total credit limit at ₹663.9 million, followed by Blue at ₹649.6 million, reflecting their larger card portfolios.

Business Recommendation

Evaluate card portfolio strategy using both average and total credit exposure. While Platinum represents higher average credit allocation per card, Gold and Blue represent greater aggregate exposure due to their larger portfolios. Credit exposure should therefore be monitored alongside card volume, customer credit profiles, and transaction behavior.

Annual Fee Analysis by Card Type

Card Type	Avg. Annual Fee
Platinum	₹1,245.17
Blue	₹1,242.16
Gold	₹1,198.56
Premium	₹1,161.66



Key Insight

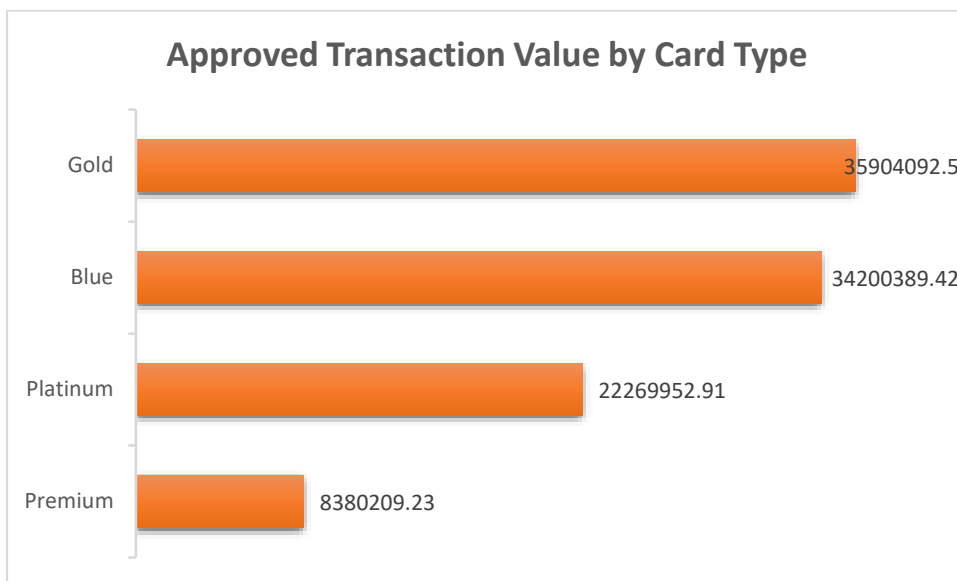
Platinum cards have the highest average annual fee at **₹1,245.17**, closely followed by Blue cards at **₹1,242.16**. Gold averages **₹1,198.56**, while Premium has the lowest average annual fee at **₹1,161.66**. The difference between the highest and lowest average fees is relatively modest at approximately **₹83.51**.

Business Recommendation

Review annual-fee positioning alongside card benefits, customer profiles, card utilization, and transaction activity to assess whether fee structures are appropriately aligned with customer engagement and card value.

Approved Transaction Value by Card Type

Rank	Card Type	Total Approved Transaction Value	Avg. Transaction Value
1	Gold	₹35,904,092.50	₹2,191.55
2	Blue	₹34,200,389.42	₹2,205.91
3	Platinum	₹22,269,952.91	₹2,260.91
4	Premium	₹8,380,209.23	₹2,231.15



Key Insight

Gold cards contribute the highest approved transaction value at ₹3.59 crore, followed by Blue cards at ₹3.42 crore. Platinum records the **highest average transaction value at ₹2,260.91**, despite ranking third in total transaction value, while Premium contributes the lowest total approved transaction value at ₹83.80 lakh.

Business Recommendation

Prioritize Gold and Blue card segments for transaction-volume and engagement analysis due to their higher aggregate approved transaction contribution. At the same time, investigate Platinum customers' higher average transaction value to understand the characteristics and behaviors associated with higher-value transactions.

TRANSACTION PERFORMANCE ANALYTICS

Monthly Approved Transaction Volume & Value Trends

Period	Transactions	Transaction Value	MoM Value Growth
Jan 2025	1,761	₹39.22L	—
Feb 2025	1,690	₹34.73L	-11.45%
Mar 2025	1,849	₹39.15L	+12.74%
Apr 2025	1,876	₹42.43L	+8.37%
May 2025	2,037	₹47.29L	+11.45%
Jun 2025	2,076	₹45.70L	-3.36%
Jul 2025	2,211	₹49.82L	+9.02%
Aug 2025	2,238	₹49.33L	-0.99%
Sep 2025	2,280	₹48.78L	-1.11%
Oct 2025	2,415	₹52.96L	+8.56%
Nov 2025	2,426	₹54.43L	+2.77%
Dec 2025	2,542	₹55.75L	+2.43%
Jan 2026	2,675	₹60.00L	+7.62%
Feb 2026	2,500	₹55.63L	-7.26%
Mar 2026	2,953	₹64.29L	+15.56%
Apr 2026	2,701	₹61.39L	-4.52%
May 2026	2,394	₹54.40L	-11.38%
Jun 2026	2,234	₹49.29L	-9.40%
Jul 2026	2,277	₹50.28L	+2.02%
Aug 2026	2,358	₹52.67L	+4.75%



Key Insight

Approved transaction activity shows an overall upward trend from early 2025 through early 2026, although several months experienced declines. Transaction value reached its highest level in March 2026 at approximately ₹64.29 lakh, accompanied by the highest monthly transaction volume of 2,953. Following declines in April–June 2026, transaction value recovered in July and August 2026, increasing 4.75% in August.

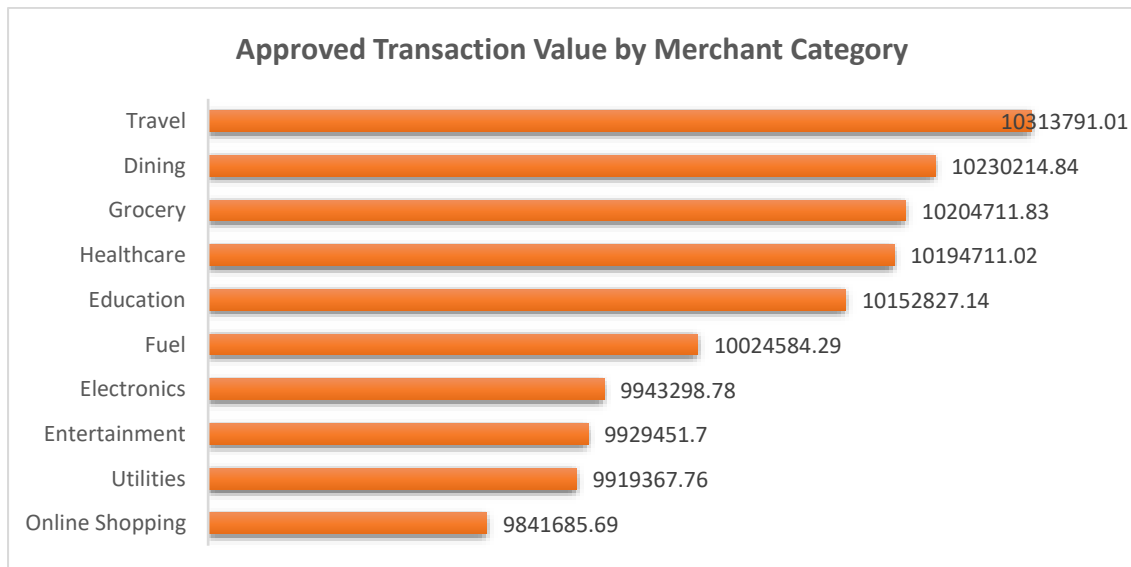
- March 2026 → **highest transaction value: ₹64.29L**
- March 2026 → **highest volume: 2,953 transactions**
- March 2026 → **highest MoM growth: +15.56%**
- May–June 2026 → notable decline
- July–August 2026 → recovery
- August 2026 → **₹52.67L**, up **4.75%** from July

Business Recommendation

Monitor monthly transaction trends to identify periods of growth and decline, and investigate the drivers behind significant changes such as the March 2026 increase and the May–June 2026 slowdown. Transaction volume and value should be analyzed alongside merchant category, card type, geography, and customer segments to identify the factors contributing to these movements.

Approved Transaction Performance by Merchant Category

Merchant Category	Transactions	Transaction Value	Avg. Transaction
Travel	4,575	₹1.031 Cr	₹2,254.38
Dining	4,675	₹1.023 Cr	₹2,188.28
Grocery	4,530	₹1.020 Cr	₹2,252.70
Healthcare	4,617	₹1.019 Cr	₹2,208.08
Education	4,516	₹1.015 Cr	₹2,248.19
Fuel	4,587	₹1.002 Cr	₹2,185.43
Electronics	4,524	₹0.994 Cr	₹2,197.90
Entertainment	4,504	₹0.993 Cr	₹2,204.59
Utilities	4,481	₹0.992 Cr	₹2,213.65
Online Shopping	4,484	₹0.984 Cr	₹2,194.85



Key Insight

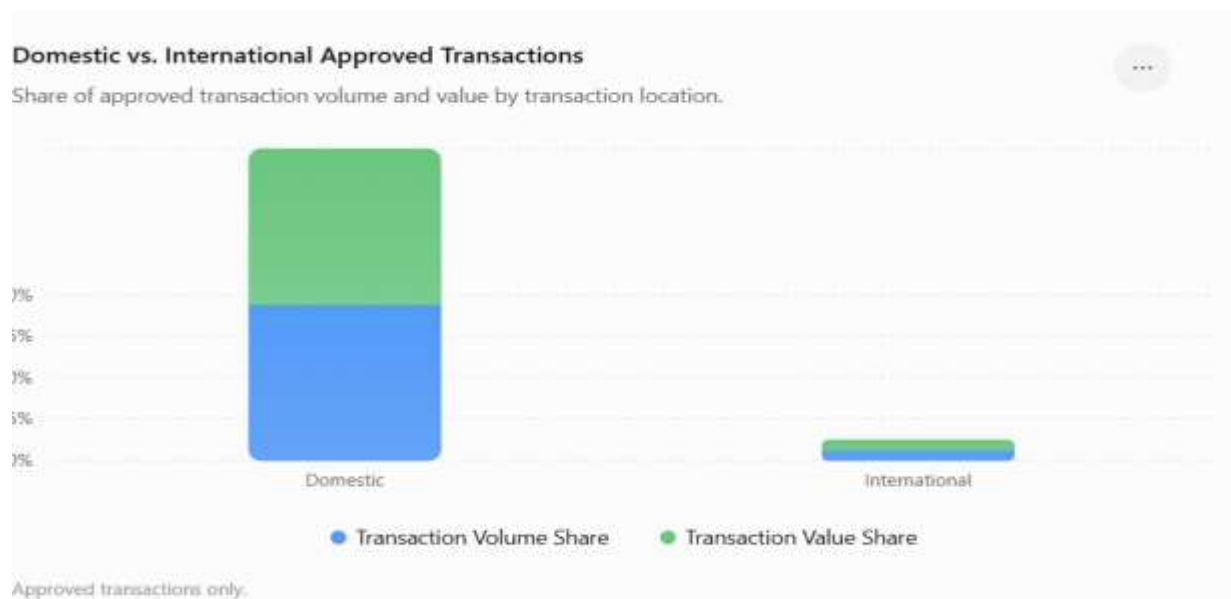
- **Travel** generates the highest approved transaction value at **₹1.031 Cr**.
- **Online Shopping** is the lowest at **₹0.984 Cr**.
- The categories are actually **quite evenly distributed**; the gap between the highest and lowest is only about **₹4.93 lakh**.
- Travel also has the highest average transaction value at **₹2,254.38**.
- Dining has the **highest transaction volume (4,675)**, despite not having the highest total value

Business Recommendation

Prioritize Travel, Dining, and Grocery for deeper transaction analysis, while investigating why Online Shopping has the lowest approved value. Combine merchant category with **card type, customer segment, geography, and transaction trends** to identify the drivers behind category-level spending.

Domestic vs. International Transaction Performance

Transaction Location	Transactions	Transaction %	Transaction Value	Value %
Domestic	42,752	93.97%	₹9.47 Cr	94.00%
International	2,741	6.03%	₹60.40 Lakh	6.00%
Total	45,493	100%	₹10.08 Cr	100%



Key Insight

- **Domestic transactions dominate**, representing **93.97% of approved transactions** and **94% of approved transaction value**.
- International transactions account for only **6.03% of volume** and **6% of value**.
- The almost identical volume and value shares suggest that domestic and international transactions have **similar average transaction values** in this dataset.

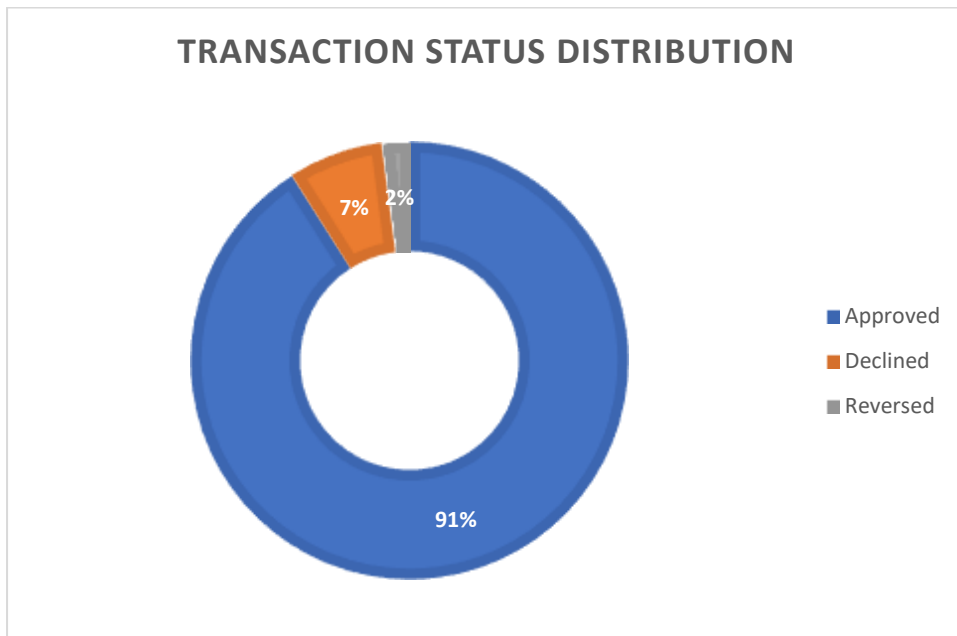
- Domestic activity is therefore the primary contributor to overall approved transaction activity

Business Recommendation

Maintain strong focus on domestic transaction activity while monitoring international transactions as a smaller but distinct segment. Further analyze international transactions by **merchant category, card type, geography, and customer segment** to identify opportunities and understand whether their lower share reflects customer behavior or portfolio characteristics.

Transaction Status Distribution

Transaction Status	Transactions	Distribution
Approved	45,493	90.99%
Declined	3,485	6.97%
Reversed	1,022	2.04%
Total	50,000	100%



Key Insight

- **90.99% of transactions were approved**, indicating that the large majority of transaction attempts were successfully processed
- **6.97% were declined**, representing the largest non-approved category
- **2.04% were reversed**, indicating a smaller portion of transactions were subsequently reversed
- Combined, **9.01% of transactions were not ultimately in the Approved status.**

Business Recommendation

Focus optimization efforts on declined transactions first, since they represent the largest non-approved category. Analyze declines by **merchant category, card type, geography, customer segment, and transaction characteristics** to identify potential drivers.

Approval & Decline Performance by Merchant Category

Merchant Category	Total Txns	Approved	Declined	Approval Rate	Decline Rate
Dining	5,150	4,675	355	90.78%	6.89%
Healthcare	5,094	4,617	369	90.64%	7.24%
Fuel	5,033	4,587	345	91.14%	6.85%
Travel	5,006	4,575	331	91.39%	6.61%
Grocery	4,976	4,530	344	91.04%	6.91%
Electronics	4,975	4,524	357	90.93%	7.18%
Education	4,977	4,516	354	90.74%	7.11%
Entertainment	4,977	4,504	375	90.50%	7.53%
Online Shopping	4,866	4,484	289	92.15%	5.94%
Utilities	4,946	4,481	366	90.60%	7.40%

Approval & Decline Rates by Merchant Category

Comparison of approved and declined transaction rates across merchant categories.



Rates are calculated against all transactions within each merchant category.

Key Insight

- **Online Shopping has the highest approval rate: 92.15%** and the lowest decline rate: **5.94%**.
- **Entertainment has the lowest approval rate: 90.50%** and highest decline rate: **7.53%**
- The approval rates are relatively close across categories, ranging from **90.50% to 92.15%**
- The decline-rate range is also relatively narrow, from **5.94% to 7.53%**.
- **Travel** also performs relatively strongly with a **91.39% approval rate**

Business Recommendation

Investigate merchant categories with comparatively higher decline rates, particularly Entertainment and Utilities, to identify potential transaction-processing or customer-behavior patterns. Online Shopping can serve as a benchmark for understanding characteristics associated with stronger approval performance.

Top Customers by Monthly Approved Transaction Spending

Period	#1 Customer	Spending	#2 Customer	Spending
Jan 2025	C105032	₹21,502.03	C100321	₹21,126.48
Feb 2025	C100503	₹25,519.69	C103696	₹18,200.81
Mar 2025	C102687	₹31,768.36	C103661	₹24,565.30
...	...	...	...	...
Mar 2026	C102476	₹62,122.18	C106334	₹42,282.42
Apr 2026	C103864	₹42,634.26	C104158	₹31,043.37
May 2026	C102924	₹42,317.58	C104558	₹33,909.99
Jun 2026	C109085	₹32,389.44	C101485	₹23,300.24
Jul 2026	C102852	₹36,132.72	C106171	₹26,803.50
Aug 2026	C100712	₹24,238.83	C107154	₹22,521.90

Monthly Top Customer Spending

Highest approved transaction spending recorded by a customer in each month.



Approved transactions only; values represent the monthly #1 customer.

Key Insight

- The highest monthly spending in this analysis occurred in **March 2026: ₹62,122.18** by **C102476**.
- October 2025 was also notably high at **₹57,986.94** by **C101272**.
- Several months show top-customer spending above ₹40,000, while August 2026 was comparatively lower at **₹24,238.83**.
- The top customer **changes considerably from month to month**, suggesting that high-value transaction activity is distributed across different customers rather than consistently concentrated in one customer.

Business Recommendation

Identify customers who repeatedly appear among the monthly top spenders and analyze their card ownership, customer profile, transaction categories, and payment behavior. This can help distinguish consistently high-activity customers from customers whose spending spikes only in particular months.

Highest-Spending Customers by Merchant Category

Merchant Category	Customer	Transactions	Approved Value	Rank
Dining	C103864	2	₹43,225.84	1
Dining	C100671	3	₹32,406.12	2
Education	C100957	2	₹30,221.98	1
Education	C107638	3	₹28,044.21	2
Electronics	C106372	1	₹31,578.08	1
Electronics	C106592	1	₹31,143.74	2
Entertainment	C102852	1	₹36,132.72	1
Entertainment	C104567	1	₹28,178.39	2
Fuel	C101032	2	₹33,771.62	1
Fuel	C108887	1	₹30,892.98	2
Grocery	C106765	3	₹50,251.66	1
Grocery	C104945	2	₹49,251.11	2
Healthcare	C104065	3	₹39,815.33	1

Healthcare	C107973	1	₹30,036.31	2
Online Shopping	C106003	1	₹47,158.86	1
Online Shopping	C102635	2	₹37,398.23	2
Travel	C102476	1	₹58,726.85	1
Travel	C103513	2	₹31,930.54	2
Utilities	C101272	2	₹58,568.00	1
Utilities	C100449	2	₹30,957.90	2

Key Insight

- **C102476** is the highest spender in Travel at **₹58,726.85**.
- **C101272** is almost identical in Utilities at **₹58,568.00**.
- **C106765** is the highest spender in Grocery at **₹50,251.66**
- **C103864** leads Dining at **₹43,225.84**.
- The highest-spending customer differs substantially across merchant categories, indicating that high-value activity is **not limited to a single customer across all categories**.

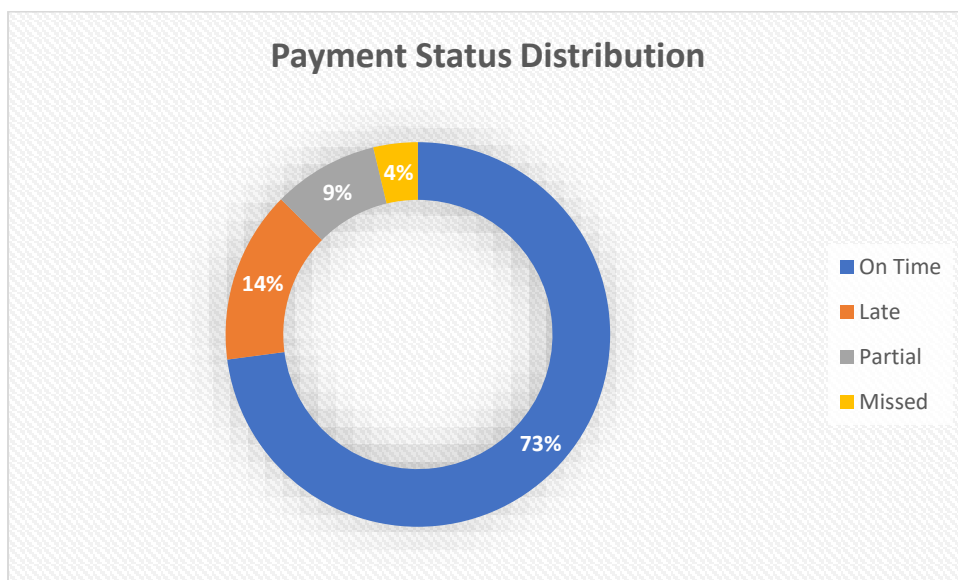
Business Recommendation

Identify customers who rank highly across multiple merchant categories and evaluate their broader transaction activity, card ownership, customer profile, and payment behavior. These customers can be treated as a high-activity segment for deeper customer intelligence and targeted engagement analysis

PAYMENT & RISK ANALYTICS

Payment Status Distribution

Payment Status	Payments	Distribution
On Time	9,479	72.92%
Late	1,883	14.48%
Partial	1,153	8.87%
Missed	485	3.73%
Total	13,000	100.00%



Key Insight

- **72.92%** of payments were made **on time**, indicating that the majority of payment activity was timely.
- **14.48%** were **late**, making late payments the largest non-on-time category.
- **8.87%** were **partial payments**.
- **3.73%** were **missed payments**.
- Overall, **27.08%** of payments were not classified as On Time.

Business Recommendation

Prioritize monitoring late, partial, and missed payments to identify payment-behavior patterns and potential areas for proactive customer engagement. Further segment these payment outcomes by card type, employment type, customer profile, and days past due to identify higher-risk patterns.

Payment Outcome & Timeliness Analysis

Payment Outcome	Number of Payments	Percentage
On Time	9,479	72.92%
Late	1,883	14.48%
Partial	1,153	8.87%
Missed	485	3.73%
Total	13,000	100.00%

Key Insight

72.92% of payments were made on time, while **14.48%** were late, **8.87%** were partial, and **3.73%** were missed. Overall, **27.08%** of payments were not classified as on time

Business Recommendation

Monitor late, partial, and missed payments to identify recurring payment-behavior patterns. Further analysis by **card type, employment type, customer profile, and days past due** can help identify segments requiring closer monitoring or targeted engagement.

Average Days Past Due by Payment Status

Payment Status	Average Days Past Due
On Time	0.00 days
Late	15.55 days
Partial	29.89 days
Missed	74.92 days



Key Insight

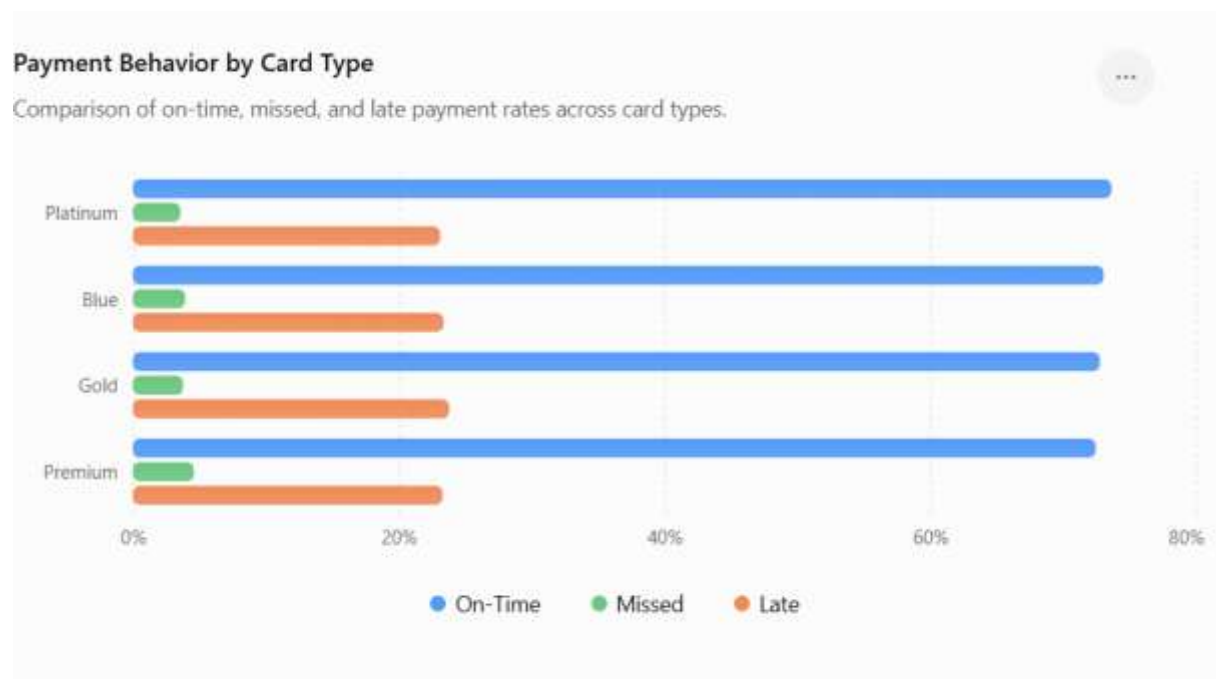
- **On Time:** 0 days past due.
- **Late:** average delay of **15.55 days**.
- **Partial:** average delay of **29.89 days**, almost twice the Late average.
- **Missed:** highest delay at **74.92 days**, indicating substantially greater payment delinquency

Business Recommendation

Monitor customers associated with partial and missed payments more closely, as these categories show substantially higher average days past due. Combine days-past-due analysis with payment frequency, customer profile, card type, and transaction activity to identify recurring payment-risk patterns.

Payment Behavior by Card Type

Card Type	On-Time Payment Rate	Missed Payment Rate	Late Payment Rate
Platinum	73.53%	3.46%	23.01%
Blue	72.94%	3.81%	23.25%
Gold	72.66%	3.65%	23.69%
Premium	72.34%	4.47%	23.19%



Key Insight

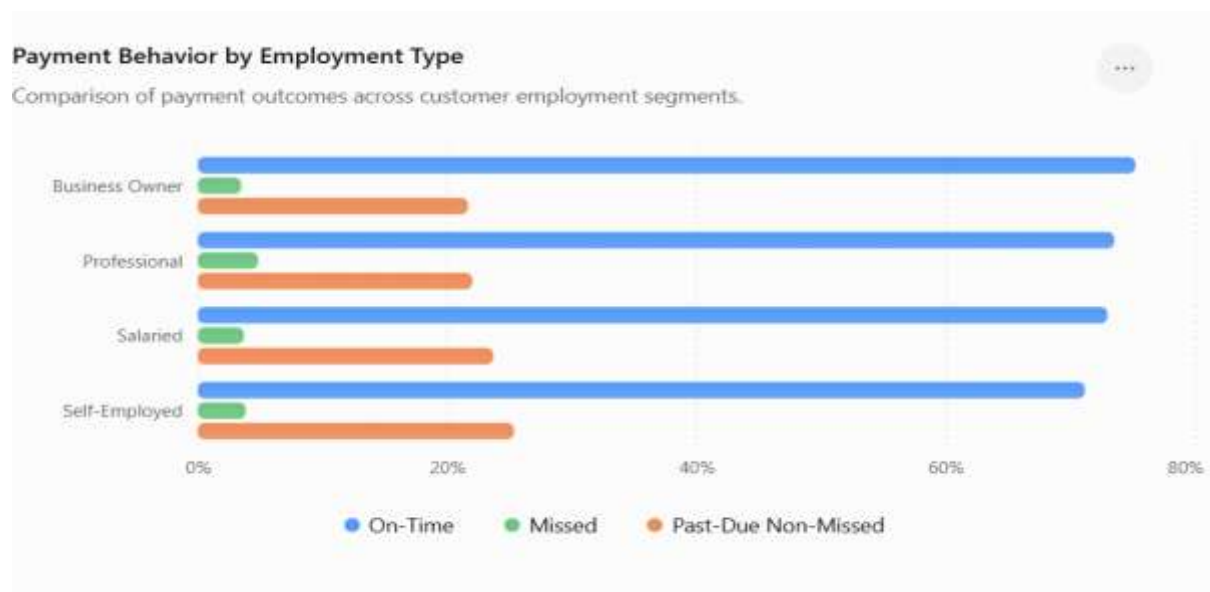
- **Platinum** has the highest on-time payment rate at **73.53%** and the lowest missed-payment rate at **3.46%**.
- **Premium** has the lowest on-time rate at **72.34%** and the highest missed-payment rate at **4.47%**.
- **Gold** has the highest late-payment rate at **23.69%**.
- Overall, payment behavior is **fairly similar across card types**, with relatively small differences in on-time rates.

Business Recommendation

Continue monitoring payment behavior across card types, with particular attention to Premium cards due to their higher missed-payment rate. Combine card-level payment behavior with customer characteristics, days past due, transaction activity, and credit profile before drawing stronger risk conclusions.

Payment Behavior by Employment Type

Employment Type	On-Time Payment Rate	Missed Payment Rate	Past-Due Non-Missed Rate
Business Owner	75.10%	3.35%	21.55%
Professional	73.40%	4.71%	21.89%
Salaried	72.86%	3.56%	23.58%
Self-Employed	71.04%	3.72%	25.24%



Key Insight

- **Business Owners** show the highest on-time payment rate at **75.10%** and lowest missed-payment rate at **3.35%**.
- **Self-Employed customers** have the lowest on-time rate at **71.04%** and highest past-due non-missed rate at **25.24%**.
- **Professionals** have the highest missed-payment rate at **4.71%**.
- The differences suggest that payment behavior varies across employment segments, particularly between **Business Owners and Self-Employed customers**.

Business Recommendation

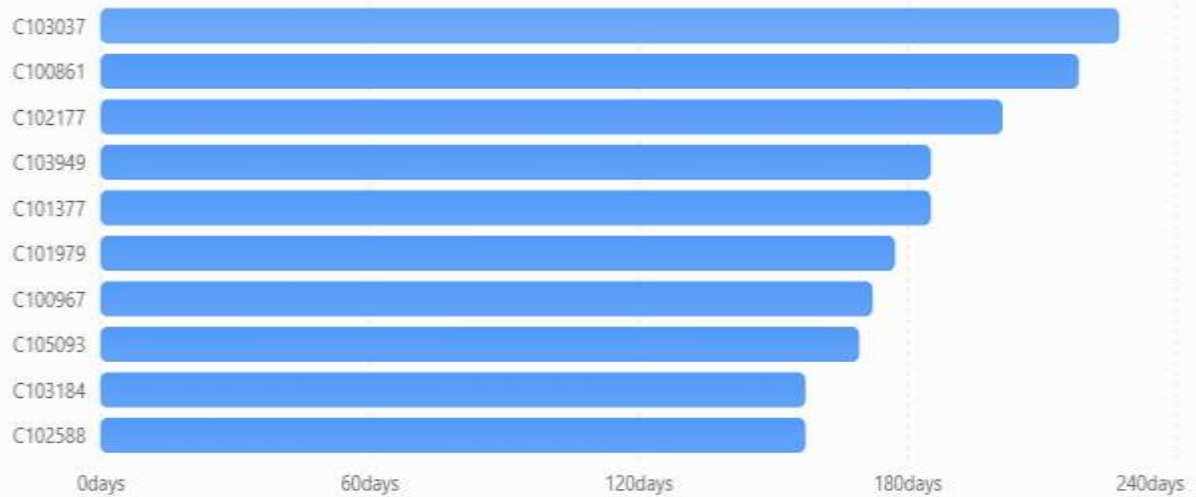
Use employment type as one segmentation dimension when monitoring payment behavior, particularly for customers showing higher past-due or missed-payment patterns. Combine employment information with days past due, credit profile, transaction activity, and historical payment behavior before making risk-related decisions.

Customers with Highest Total Days Past Due

Rank	Customer ID	Total Days Past Due
1	C103037	227 days
2	C100861	218 days
3	C102177	201 days
4	C103949	185 days
5	C101377	185 days
6	C101979	177 days
7	C100967	172 days
8	C105093	169 days
9	C103184	157 days
10	C102588	157 days

Top 10 Customers by Total Days Past Due

Customers with the highest cumulative days past due across payment records.



Key Insight

- **C103037** has the highest cumulative past-due duration at **227 days**.
- The top 10 customers range from **157 to 227 total days past due**.
- Several customers have accumulated substantial payment delays, making them useful candidates for deeper payment-behavior analysis.

Business Recommendation

Prioritize customers with high cumulative days past due for further review of their payment history. Analyze the frequency and severity of their late, partial, and missed payments to distinguish isolated delays from recurring payment-behavior patterns.

Key Business Questions

1. Customer Intelligence

Q1. Which customer segments generate the highest transaction value?

The analysis shows that **Gold cardholders generated the highest approved transaction value at ₹3.59 crore**, followed by **Blue cardholders at ₹3.42 crore**. At the customer level, several customers also demonstrated high transaction activity, with the top customer generating **₹72,785.69** in approved transaction spending.

Answer: Gold and Blue card segments contribute the largest approved transaction value, while high-value transaction activity is distributed across multiple individual customers.

Q2. How does income and credit profile relate to customer spending behavior?

Average income was relatively similar across employment groups, ranging from approximately **₹8.55 lakh to ₹8.67 lakh**. Average credit scores were also tightly grouped between **717.50 and 718.99**.

The analysis therefore does **not show a strong difference in income or average credit score across employment groups**. These characteristics should be evaluated together with transaction activity rather than independently.

Answer: Income and average credit score show limited variation across employment segments, suggesting that **income or employment alone should not be used to explain spending behavior**. Combining these factors with transaction activity, card ownership, and payment behavior provides a more meaningful customer profile

Q3. Which customers demonstrate high-value or highly active behavior?

The highest approved-spending customer was **C101272 with ₹72,785.69**, followed by **C103864 with ₹69,424.75** and **C102476 with ₹65,402.73**.

The monthly analysis also showed that different customers emerged as top spenders across different months and merchant categories.

Answer: High-value transaction activity is distributed across multiple customers, with several customers demonstrating significantly higher approved spending than the broader customer base.

2. Card Portfolio

Q4. Which card types have the highest customer adoption and transaction activity?

Gold and Blue have the largest card portfolios:

- Gold: **4,301 cards**
- Blue: **4,148 cards**
- Platinum: **2,586 cards**
- Premium: **965 cards**

Gold also generated the highest approved transaction value at **₹3.59 crore**, followed by Blue at **₹3.42 crore**.

Answer: Gold and Blue have the strongest portfolio presence and contribute the largest approved transaction values.

Q5. How does credit-limit distribution vary across card types?

Platinum had the highest average credit limit at **₹156,670.53**, while Blue had the highest total credit exposure at **₹64.96 crore** because of its larger card portfolio.

Card Type	Avg Credit Limit	Total Credit Limit
Platinum	₹1,56,670.53	₹40.52 Cr
Blue	₹1,56,605.59	₹64.96 Cr
Gold	₹1,54,359.45	₹66.39 Cr
Premium	₹1,50,932.64	₹14.57 Cr

Answer: Platinum has the highest average credit limit, while Gold has the highest total credit exposure due to its larger portfolio size.

Q6. Which card categories contribute most to overall transaction value?

Gold contributed the highest approved transaction value at **₹35.90 million (₹3.59 crore)**, followed by Blue at **₹34.20 million (₹3.42 crore)**.

Answer: Gold and Blue are the largest contributors to approved transaction value, together accounting for approximately **69.6%** of total approved transaction value.

3. Transaction Analytics

Q7. What are the major transaction categories and spending patterns?

Transaction value was relatively evenly distributed across merchant categories

Travel had the highest approved transaction value at **₹1.031 crore**, while **Online Shopping** had the lowest at approximately **₹0.984 crore**.

Dining had the highest approved transaction volume with **4,675 transactions**.

Answer: Travel generated the highest transaction value, while Dining recorded the highest transaction volume. Overall spending was broadly distributed across merchant categories rather than being concentrated in one category.

Q8. How does transaction activity change over time?

Approved transaction activity generally increased from early 2025 into early 2026, although there were several month-to-month declines.

March 2026 recorded the highest approved transaction volume at **2,953 transactions** and the highest approved transaction value at approximately **₹64.29 lakh**.

After declines during May and June 2026, activity recovered in July and August.

Answer: Approved transaction activity showed an overall upward pattern with periodic fluctuations, reaching its highest point in **March 2026**, followed by a mid-2026 slowdown and subsequent recovery.

Q9. Which customer, card, or merchant segments contribute the most transaction value?

At the card level, **Gold** contributed the highest approved transaction value (**₹3.59 crore**).

At the merchant level, **Travel** generated the highest approved transaction value (**₹1.03 crore**).

At the customer level, **C101272** had the highest overall approved spending among the analyzed customers (**₹72,785.69**).

Answer: Gold cards, Travel transactions, and a group of high-value customers are the strongest contributors identified by the analysis

Q10. What proportion of transactions are approved, declined, or reversed?

Transaction Status	Transactions	Percentage
Approved	45,493	90.99%
Declined	3,485	6.97%
Reversed	1,022	2.04%
Total	50,000	100%

Answer: 90.99% of transactions were approved, 6.97% were declined, and 2.04% were reversed.

4. Payment & Risk

Q11. What proportion of payments are made on time, late, partially, or missed?

Payment Status	Payments	Percentage
On Time	9,479	72.92%
Late	1,883	14.48%
Partial	1,153	8.87%
Missed	485	3.73%
Total	13,000	100%

Answer: 72.92% of payments were on time, while 14.48% were late, 8.87% were partial, and 3.73% were missed.

Q12. Which customer segments show higher payment delays?

Employment Type	On-Time Rate	Past-Due Non-Missed Rate
Business Owner	75.10%	21.55%
Professional	73.40%	21.89%
Salaried	72.86%	23.58%
Self-Employed	71.04%	25.24%

Answer: Self-Employed customers show the highest past-due non-missed rate at **25.24%** and the lowest on-time payment rate at **71.04%** among the employment segments analyzed.

Q14. How does payment behavior vary by card type or customer profile?

Payment behavior was relatively similar across card types:

- Platinum had the highest on-time rate: **73.53%**
- Premium had the lowest on-time rate: **72.34%**
- Premium had the highest missed-payment rate: **4.47%**
- Gold had the highest past-due non-missed rate: **23.69%**

By employment type, Business Owners had the highest on-time rate (**75.10%**), while Self-Employed customers had the highest past-due non-missed rate (**25.24%**)

Answer: Payment behavior varies across both card and employment segments, although the differences are relatively moderate. Employment type shows a somewhat larger spread than card type in the analyzed payment rates.

FINAL BUSINESS INSIGHTS

1. Strong Transaction Approval Performance

The portfolio recorded a **90.99% transaction approval rate**, with 45,493 of 50,000 transactions approved. Declined transactions represented **6.97%**, while reversals accounted for **2.04%**.

This indicates that the majority of transaction attempts were successfully processed, while declined and reversed transactions remain areas for further investigation.

2. Gold and Blue Cards Drive the Largest Transaction Value

Gold cards generated approximately **₹3.59 crore** in approved transaction value, followed by Blue cards at approximately **₹3.42 crore**.

Together, these two card types contributed approximately **69.6% of total approved transaction value**.

This makes Gold and Blue important segments for understanding transaction activity and customer engagement.

3. Customer Spending Is Distributed Across Merchant Categories

Travel generated the highest approved transaction value at approximately **₹1.03 crore**, while Dining recorded the highest approved transaction volume with **4,675 transactions**.

At the same time, transaction values across merchant categories were relatively close, indicating that spending was **not heavily concentrated in a single merchant category**.

4. Transaction Activity Reached Its Highest Point in March 2026

Approved transaction activity generally increased from early 2025 into early 2026, although monthly fluctuations were observed.

March 2026 recorded the highest approved transaction volume at **2,953 transactions** and the highest approved transaction value at approximately **₹64.29 lakh**.

Activity declined during May and June 2026 before recovering in July and August.

This suggests an opportunity to investigate the factors behind significant monthly changes in transaction activity.

5. The Customer Base Is Predominantly Single-Card

81.79% of customers own one card, while **16.47% own two cards**. Only approximately **1.74%** own three or more cards.

This indicates that the customer base is largely concentrated around single-card ownership, while multi-card customers represent a relatively small segment.

6. Payment Timeliness Is an Important Monitoring Area

Only **72.92% of payments were made on time**.

The remaining payments consisted of:

- **14.48% Late**
- **8.87% Partial**
- **3.73% Missed**

Therefore, approximately **27.08% of payments were not classified as on time**.

This makes payment behavior an important area for customer-level monitoring and segmentation.

7. Payment Delay Severity Increases With Payment Outcome

Average Days Past Due differed substantially by payment status:

- On Time: **0 days**
- Late: **15.55 days**
- Partial: **29.89 days**
- Missed: **74.92 days**

The increasing DPD pattern provides a useful indicator for distinguishing different levels of payment difficulty.

8. Payment Behavior Varies Across Customer Segments

Self-Employed customers recorded the lowest on-time payment rate at **71.04%** and the highest past-due non-missed rate at **25.24%** among the employment groups analyzed.

Business Owners had the highest on-time rate at **75.10%**.

However, these differences should be interpreted together with credit profile, transaction behavior, and payment history rather than using employment type alone to assess risk.

9. A Small Group of Customers Shows High Cumulative Payment Delays

The customer-level DPD analysis identified customers with substantial cumulative days past due.

For example:

- C103037 — **227 days**
- C100861 — **218 days**
- C102177 — **201 days**

These customers represent candidates for **deeper payment-history review**.

Importantly, cumulative DPD alone should not be interpreted as proof of repeated missed payments or as a standalone measure of credit risk.

10. Credit Profiles Are Broadly Concentrated in the Middle-to-Upper Ranges

67.36% of customers were classified as having **Good or Very Good** credit scores.

Only **1.61%** were classified as Poor.

This indicates that the customer portfolio is primarily concentrated in Good and Very Good credit-score categories, although credit score should be evaluated together with actual payment behavior and transaction activity.

BUSINESS RECOMMENDATIONS

Area	Business Recommendation
Transaction Performance	Analyze declined transactions by merchant category, card type, geography, customer segment, and time period to identify potential improvement opportunities.
High-Value Customers	Identify customers with consistently high transaction activity and evaluate their card ownership, spending patterns, and payment behavior for targeted engagement.
Card Portfolio	Use Gold and Blue transaction activity as a benchmark while analyzing why Platinum and Premium customers exhibit different transaction and payment patterns.
Customer Engagement	Analyze single-card customers to identify suitable cross-sell opportunities based on transaction behavior, credit profile, and customer characteristics.
Merchant Analytics	Investigate high-value categories such as Travel and high-volume categories such as Dining to understand customer preferences and transaction drivers.
Payment Monitoring	Monitor Late, Partial, and Missed payments using Days Past Due, payment frequency, and historical payment behavior.
Customer Segmentation	Combine employment type, income, credit score, card ownership, transaction activity, and payment behavior instead of relying on a single customer attribute.
High-DPD Customers	Conduct deeper payment-history reviews for customers with consistently high cumulative Days Past Due.
Transaction Trends	Investigate the drivers behind the March 2026 peak and the May–June slowdown to understand changes in customer transaction activity.
Risk Analytics	Develop customer-level monitoring using multiple behavioral indicators rather than treating credit score or employment type as standalone risk measures.

FINAL CONCLUSION

This project demonstrates how **customer, card, transaction, and payment data can be integrated to generate actionable financial and customer intelligence.**

The analysis covered **10,000 customers, 12,000 cards, 50,000 transactions, and 13,000 payments**, using **Excel, Power Query, MySQL, SQL, and Power BI** to perform data validation, analysis, segmentation, and reporting.

The analysis identified several important patterns, including strong transaction approval performance, significant transaction contribution from Gold and Blue cards, variation in merchant-category activity, concentration of customers around single-card ownership, and meaningful differences in payment behavior across customer segments.

The payment analysis also highlighted the importance of combining **Payment Status and Days Past Due** with customer and transaction characteristics when monitoring payment behavior.

Overall, the project demonstrates an end-to-end analytical workflow:

Data Validation → SQL Analysis → Business Questions → Insights → Recommendations → Business Intelligence

The key strength of the project is not simply the number of SQL queries, but the ability to **translate financial data into structured business questions, evidence-based insights, and practical recommendations.**